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2.1 Data Representation

2.2 Statistical Data Analysis

2.3 Data Visualization

2.4 Dimensionality Reduction

2.5 Data Classification

2.6 Data Prediction

Video: Data Classification

1h 39m

Graded App Item: my_fitpca

Submitted

Graded App Item: Mahalanobis Distance

Submitted

Graded App Item: my_predictpca

15 min

Quiz: Smartphone activity classifier

Started

🎉 Congratulations! You passed!

Grade received 100%

Latest Submission Do not resubmit

Grade 100%

Go to next item

Smartphone activity classifier

Review Learning Objectives

1. In the previous Matlab grader problems, we have created our own classifier method based on fitting PCA models to different classes. Now let's try it out on a new dataset! 1 / 1 point

The humanactivity dataset contains thousands of samples of human activities, 60 feature dataset. The 'class' is 1 of 5 possible human activities: sitting, walking, dancing, running, and standing. The 'features' are features derived from acceleration measurements from the accelerometer sensor of a smartphone located in the participant's pocket taken during the course of the recorded activity. If interested, you can find more information about this dataset [here](#). [?]

Try again

Receive grade

Your goal is to create a classifier program that could be run on a smartphone to classify a person's activity from sensor measurements.

Your grade 100%

View Feedback

We keep your highest score

- Load the humanactivity dataset into MATLAB and divide it into a training and testing set using the following code.

```
1 load humact.mat
2 D_train = feat(1:2:end,:);
3 class_train = actid(1:2:end);
4 D_test = feat(2:2:end,:);
5 class_test = actid(2:2:end);
```

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- Use your my_fitpca function create a classification model with the training set.
- Use your classification model with your my_predictpca function to estimate classifications on the testing set

What is your classification accuracy on the testing set (percent correct), accurate to 2 decimal places)?

humact
MAT File

96.73

Correct